

Syntactic Simplification for Improved Text-to-Pictograph Translation

CLIN27

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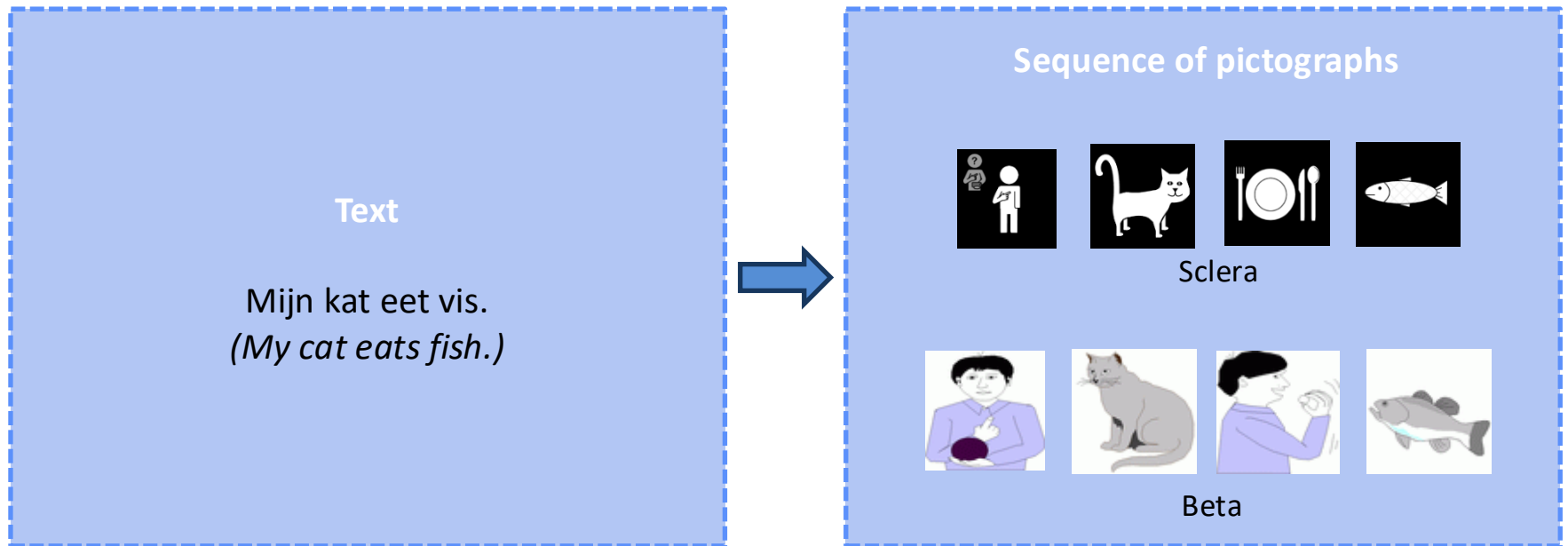


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Introduction

- **Text-to-Pictograph** translation
- To make the internet accessible for people with **Intellectual Disabilities**



Problem Definition

Long input sentences

Volgend jaar in september willen we samen basket doen in de sporthal van de school.

Next year in September want we together basket do in the sports-hall of the school.

‘Next year in September we want to play basket together in the sports hall of the school.’



Coordinated and complex sentences

Ik heb de tekening die jij aan mij gegeven hebt verscheurd.

I have the drawing that you to me given have torn-up.

'I have torn up the drawing that you gave me.'



Non-SVO order

Hij zegt dat hij ons een berichtje gemaild heeft.

He says that he us a message mailed has.

'He says that he has mailed us a message.'



Passive sentences

De postbode wordt gebeten door de hond.

The mailman is bitten by the dog.

‘The mailman is bitten by the dog.’



Solution:

Apply syntactic simplification as a pre-processing step
for Text-to-Pictograph translation



Status Quaestionis

- Syntactic simplification (vs. lexical simplification)
- **Rule-based systems without parsers**
 - *Chandrasekar et al. 1996*
 - These systems often raised more issues than they addressed
- **Rule-based systems with parsers**
 - *Carroll et al. 1999; Candido Jr. et al. 2009; Bott et al. 2012*
 - Limited in scope to lexical simplification
 - Strong syntactic simplifications
 - In some cases: bottleneck = time (reparsing until all constructs are simplified)
- **Machine Translation approaches**
 - Use phrase tables (needs large parallel corpora)
 - Not well suited for splitting or reordering operations
- **Compression**
 - *Daelemans et al. 2004; Vandeghinste & Pan 2004*
 - $\neq$ Syntactic simplification



- **Our approach:**
 - Machine Translation is **not** appropriate
 - No parallel data (Dutch – simplified Dutch)
 - Cannot perform strong simplifications (needed for our target users)
 - Rule-based systems
 - Do not require parallel data
 - Old systems: parsers would time out, not reliable
 - **Today: fast and reliable parsing**
 - Compression
 - Useful for users who are still getting used to the pictograph language
 - ! Not the same as syntactic simplification, so extra module



Objectives

ID	Phenomenon	Operations
(a)	Coordinated sentence	- Split into two or multiple sentences - Identify the antecedent (subject) if subject is covert (in case of ellipsis)
(b)	Subordinate clause	- Detach from the main clause
(c)	Clause order	- Determine the subordinate clause's temporal or logical position with respect to the main clause
(d)	Participial phrase	- Detach from the main clause - Identify the antecedent (subject)
(e)	(om) te "to" + infinitive clause	- Detach from the main clause - Identify the antecedent (subject)
(f)	Relative clause	- Detach from the main clause - Identify the antecedent (subject, direct object, or indirect object)
(g)	Embedded appositive clause	- Place the verb <i>to be</i> in front of the apposition - Identify the antecedent (subject)
(h)	Adverbial phrase or prepositional phrase in theme position	- Move to the back of the sequence
(i)	Covert subjects or objects in the newly created independent clause	- Place all the antecedents at their appropriate positions within the clause
(j)	Non subject-verb-object order	- Convert to subject-verb-object order - Cluster all the verbs at the verb position
(k)	Passive voice	- Place the agent at the patient's position and vice versa
(l)	Extra: Compression	- Delete pictographs that do not contribute to the essence of the message

Ik ben woensdag een lieve baby gaan bezoeken en ik heb hem een papfles gegeven.
 I am Wednesday a cute baby go visit and I have him a bottle given.
 'On Wednesday I went to visit a cute baby and I gave him the bottle.'



Old



New

De vogel die jij hoort is een pauw.
The bird that you hear is a peacock.
'The bird that you're hearing is a peacock.'

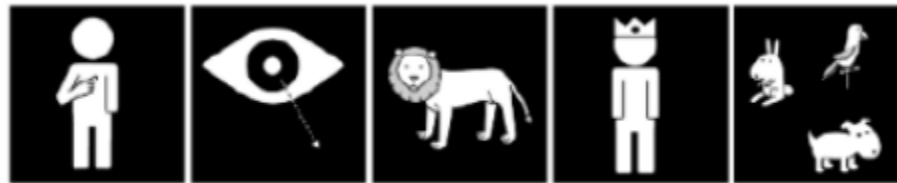


Old



New

Ik zag een leeuw, de koning van de dieren.
I saw a lion, the king of the animals.
'I saw a lion, the king of the animals'



Old



New

Ik moet douchen nadat we gegeten hebben.

I must shower after we eaten have.

‘I must take a shower after we have eaten.’



Old



New

Het liedje wordt gezongen door de juf.
The song is sung by the teacher.
'The song is sung by the teacher.'



Old



New

System Description

1. **Applying Alpino** (*van Noord, 2011*)
2. **Processing the Alpino XML treebank**
 - Create a sentence object for every clause
 - Mark clause type, indicate interrogation and passivity, identify the antecedents
3. **Inserting the antecedents and re-ordering the syntactic constituents**
 - Detopicalise prepositional phrases and adverbs
 - Create a verb cluster
 - Create SVO-type clauses (insert antecedent at its S or O position, move verb cluster to V position)
 - Appositive clauses: add a *to be*
 - “Passive” feature: switch agent and patient, remove *worden* “to be”



De postbode, die gisteren door onze hond gebeten werd,
The mailman, who yesterday by our dog bitten was,

eist dat we een nieuwe broek voor hem kopen.
demands that we a new pair-of-pants for him buy.

"The mailman, who was bitten by our dog last week, demands that we buy him a new pair of pants."

De postbode eist
The mailman demands



(b)

die gisteren door onze hond gebeten werd
who yesterday by our dog bitten was
(Antecedent: postbode "mailman", SUBJ)

(h) - (i) - (j)

postbode gebeten werd door onze hond gisteren
mailman bitten was by our dog yesterday

dat we een nieuwe broek voor hem kopen
that we a new pair-of-pants for him buy

(i)

(dat) we kopen een nieuwe broek voor hem
(that) we buy a new pair-of-pants for him



(k)

hond gebeten postbode gisteren
dog bitten mailman yesterday



Evaluation

	De Standaard		WAI-NOT	
	Cases Correct		Cases Correct	
Split coordinated sentences	37	36	97	90
Detach subordinate clauses	51	50	18	17
Change clause order (when temporality or a logic relation is expressed)	1	1	1	1
			0	0
Detach participial phrases	0	0	3	3
Detach (<i>om</i>) <i>te</i> “to” + infinitive clauses	25	25	0	0
Detach relative clauses	31	31	0	0
Detach appositive clauses	16	15	0	0
Move adverbial clauses or Prepositional Phrases in theme position to the back	27	26	4	4
Insert any covert subjects or objects in the newly created independent clause, at the correct position	89	73	3	3
Convert non subject-verb-object order into subject-verb-object order	153	147	49	49
Convert passive voice into active voice	9	9	1	1



Conclusions

- **First step toward the creation of a simplification tool for Dutch**
 - I.e., for generating textual output, not pictograph output
 - Requires additional grammar checker
 - Requires additional lexical simplification
- **Fast: only one syntactic parse (Alpino) needed per sentence**
 - Using recursion and loops
 - Applying the simplification rules in a logical way
- **To do: human judgments (Able to Include)**





Thank you!

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Picto demo & publications:

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